Study Guide for Exam #2 with Practice Problems

MATH 241 — Probability, Fall 2026

General Information

The second exam will be held in-class on Tuesday, November 10th. Keep in mind the following things:

- You will have 75 minutes to complete the exam, unless you have an accommodation.
- Please bring a calculator!!
- This is a **closed-book** exam, with the exception of **two** 8.5x11" sheets of paper. You can use both sides of the sheets to write down anything and you can re-use the sheet from the last exam if you so desire for one of the two sheets.
- There will be minimal R code on this exam. R code will only be provided to you to complete questions; you will not have to write any R code.
- Cell-phones, laptops, calculators, and any other electronic devices are not permitted. No interaction with anyone is allowed.

Expect the exam to contain about 4-5 questions, with a few subparts to each question.

Topics

Everything covered from October 1st to November 3rd is considered fair game for the exam. Additionally, while concepts from the first two chapters will not be covered explicitly on this exam, you might find that those concepts might pop up when solving problems related to recent material (e.g. complement rule for solving upper-tail probabilities).

These will be the main topics covered on the exam:

0. Ch. 3 (3.1-3.8): Random Variables
    - Identifying and writing functionality, notation
    - Defining, verifying, and using probability mass functions
    - Identifying common distributions and their parameters, such as the Binomial, Geometric, Discrete Uniform, and Poisson

- Defining and determining cumulative density functions using summation
- Connecting independence and functions of RVs to the aforementioned items
- Solving for point and cumulative probabilities using the PMF and CDF
- Identifying R code used to aid in solving for discrete probabilities

1. Ch. 4 (4.1, 4.2, 4.4-4.6): Expectation
   - Finding expectations, variances, and the first two moments by definition or through LOTUS
   - Defining expectation for any random variable
2. Ch. 5 (5.1-5.5): Continuous Random Variables
   - Defining and using probability density functions
   - Finding CDFs using integration of PDFs
   - Finding PDFs using derivatives of CDFs
   - Identifying common distributions and their parameters, such as the Normal, Exponential, and Uniform
   - Finding expectations and variances for continuous RVs
   - Solving for probabilities using the CDF and PDF
   - Identifying R code used to aid in solving for continuous probabilities
3. Parts of Ch. 6 (6.1-6.2): Moments
   - Finding summaries in a distribution
   - Identifying which moments correspond to which distribution characteristics

Tips

As mentioned in the first study guide, practice makes perfect, and especially so in probability. The textbook has many problems that have available solutions online via https://stat110.net. However, these questions may extend beyond what is required for the exam – stick to the objectives I've mentioned above to determine which parts to work on for the textbook problems.

Ultimately, focus on trying to find the following items for discrete and continuous random variables:

- PMF/PDF $f_X(x)$
- CDF $F_X(x)$
- Expectation $E(X)$

- Variance $Var(X)$
- First two moments $E(X)$, $E(X^2)$
- Probabilities $P(X = x)$, $P(X \leq x)$, $P(X \geq x)$, $P(a \leq X \leq b)$

Similar to previous problems, take the time to write down everything you know about the random variable, which may include the above items, parameters, and events you may be looking for.

You may find a LLM useful for developing additional practice problems. For instance, you can prompt a LLM with something like this:

> Generate practice questions in reference to finding the CDF, expectation, and variance of various continuous random variables, such as non-common and common distributions.

While some questions may not be worth doing, most of them will be in the same vein of what you may see on the exam.

Lastly, make the most of the resources you have. I will have three different office hour time blocks throughout the next week, and we will have a review session on Thursday. Come to Thursday's class with any clarifying questions you would like answered.

Practice Problems

I've provided some practice problems below. You will work on some of these problems as part of HW7.

1. Let X be a random variable, which has the following PDF:

$$f_X(x) = 8x\, e^{-4x^2} \quad \text{for } x \geq 0$$

   (a) Find the CDF of X.

   (b) Find $P(X > 1)$.

   (c) Find x such that $P(X \leq x) = 0.75$.

2. Suppose X has the following PDF:

$$f_X(x) = c\, e^x \quad \text{for } 2 < x < 4$$

   (a) Find c. *Hint*: What is the value of the integration of $f_X(x)$ from 2 to 4?

   (b) Using part (a), find the CDF.

   (c) Find $P(X < 3)$ and $P(X > 3)$.

3. Let X be a random variable with the following CDF:

$$F_X(x) = \frac{1}{\pi} arctan\left(\frac{x}{2}\right) + \frac{1}{2} \quad \text{for } -\infty < x < \infty$$

   (a) Find the PDF of X.

   (b) Find $E(X)$.

   (c) Find $P(0 < X < 2)$.

4. Suppose X represents the lifespan of a heater (in years), which has the following PDF:

$$f_X(x) = \frac{1}{4}\, e^{-\frac{1}{4}(x-5)} \quad \text{for } x \geq 5$$

   (a) Find $E(X)$ and $Var(X)$.

   (b) Sketch the PDF of X for $x \in [0, 10]$.

   (c) Suppose you buy this heater twice, one after another heater dies. Discuss if having these two heaters last about 16 years will be worth it compared to the norm.

5. Suppose X has the following PMF:

$$f_X(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \text{for } x = 0, 1, 2, \ldots$$

(a) Show that this PMF is a valid PMF.

(b) Set $\lambda = 2$. Find $E(X)$ using summation.

(c) Find $P(X > 2)$ with $\lambda = 2$.

6. Suppose Caleb is in search of an ultra-rare baseball card, in which for any card, there is a 0.1% chance of getting that card. Define X as the number of cards that are ultra-rare within a pack. There are 5 cards in a pack.

(a) Write the PMF of X.

(b) Determine the chance of getting an ultra-rare card from a pack.

(c) Let Y be the first **pack** that has an ultra-rare card. Find the expectation of Y.

7. Suppose Jamie wants to buy 4 new plants for her apartment, but is currently indecisive about which kinds of plants she wants. She visits a nursery and finds that they have 10 monsteras (3 of which are small, 4 medium, and 3 large). The plants have the following pot sizes:

- Small: 4″ diameter
- Medium: 9″ diameter
- Large: 12″ diameter

Let X be the pot size of any randomly selected monstera plant in this nursery.

(a) Write the PMF for X.

(b) Find the average pot size of all monsteras.

(c) Say Jamie wants to buy a plant and place it into a new pot that is 1.2 times the size of the original pot plus 1 inch. Find the average new pot size.

8. On any given day, a household will receive a certain number of pieces of mail, denoted as X. The distribution is as follows:

x	0	1	2	3	4
Prob.	0.17	0.05	0.17	0.24	0.37

(a) Find the CDF of X.

(b) Find $P(1 < X < 4)$ and $P(X \leq 1)$.

(c) Let $Y = I(X \geq 3)$.

(i) Find the distribution of Y and its expectation.

(ii) Discuss the meaning of Y.

9. Create your own discrete or continuous random variable based off some real-life scenario. For this random variable, find:

(a) its PDF/PMF.

(b) its CMF.

(c) its expectation and variance.

(d) the first, second, and third quartiles.

10. Consider some insurance policy, where there is a $1/100$ chance that you pay out \$1 million. There is a $1/10$ chance that you pay out \$100,000. There is a 39% chance that you pay out \$10,000. No more than one of these scenarios can happen. The remaining 50% of the time, you pay out nothing. Let X be the payout of this policy. The expected payout is $E(X) = 23,900$.

(a) Find the variance and standard deviation of the payout of this policy.

(b) If you sell the policy for its expected value, what is the probability that you will lose money?

(c) Suppose you sell two independent policies. What is the expectation, variance, and standard deviation for the sum of the payouts from the policies?

(d) With two independent policies, each sold for the expected payout value, what is the probability that you will lose money?